COURSE OUTLINE

Gennoor Tech
TRAIN. INNOVATE. BUILD.

Agentic AI Vulnerability Management

From scanner noise to prioritized, verified remediation

3 days (8 hours per day) | Intermediate | Instructor-led, hands-on labs | Capstone project

About this course

Who should attend

Vulnerability management, AppSec, IT operations, security engineering and remediation owners.

Prerequisites

- Basic vulnerability concepts (CVE, CVSS, scanners)
- JSON and API familiarity
- Python fundamentals
- Issue-tracking concepts (Jira, ServiceNow or similar)

What you will be able to do

- Normalize findings from one or more scanner formats into a stable schema with traceable deduplication
- Correlate findings with asset criticality, exposure and evidence quality
- Create explainable prioritization that separates facts from inferences
- Generate actionable, idempotent remediation tickets with ownership and due-date logic
- Verify remediation through rescan or configuration evidence rather than ticket status

Tools and technologies

Claude | Python | FastAPI | Pydantic | PostgreSQL | Jira (sandbox) | ServiceNow (mock) | pytest | Docker Compose | Trivy | Snyk

Gennoor Tech TRAIN. INNOVATE. BUILD. For more info - https://gennoor.com

Table of contents

DAY 1 3 modules

MODULE 1

The finding pipeline: data model and schemas

- Finding, asset, evidence, remediation, ticket and verification schemas
- Source, timestamps, affected version, severity and confidence as first-class fields
- Shared engineering principles and the AI-assisted development workflow

HANDS-ON LAB
Design the normalized schema with the coding assistant; review migration risks before writing code.

DELIVERABLE
Agreed data model.

MODULE 2

Import, normalize and deduplicate

- One scanner export format first; validation and import summary with rejected rows
- Deterministic deduplication keys and traceable duplicate links
- Data quality checks

HANDS-ON LAB
Build the importer and prove duplicate reduction on a seeded dataset.

DELIVERABLE
Repeatable normalized finding database.

MODULE 3

Asset context and business criticality

- Joining findings to synthetic CMDB data: owner, criticality, internet exposure, environment, dependencies
- Missing context is reported, never invented

HANDS-ON LAB
Attach asset context and surface findings with no owner.

DELIVERABLE
Context-enriched findings.

DAY 2 3 modules

GenNoor Tech TRAINING INNOVATE LEARN
For more info - https://gennoor.com

MODULE 4

Threat and exploit intelligence enrichment

- Adapters for advisory and exploit-intelligence data
- Curated local datasets for deterministic labs
- Evidence quality grading

HANDS-ON LAB
Enrich findings from the local intelligence set and grade the evidence.

DELIVERABLE
Enrichment adapters.

MODULE 5

Explainable prioritization with an agent

- Transparent baseline score or rules; the model explains, it does not override policy
- Agent prompt that cites evidence, separates facts from inferences and returns unknown when evidence is insufficient
- Strict output schema and tests for missing context, conflicting severity and stale data

HANDS-ON LAB
Implement the prioritization engine and the explanation agent; test it with adversarial inputs.

DELIVERABLE
Explainable prioritized backlog.

MODULE 6

Remediation playbooks and ticket orchestration

- Mapping finding types to approved remediation playbooks
- Ticket fields, acceptance criteria, ownership, SLA and due-date logic
- Idempotent create and update against a mock or sandbox ticket system

HANDS-ON LAB
Generate remediation tickets in the sandbox and re-run the pipeline to prove idempotency.

DELIVERABLE
Ticket workflow in sandbox.

DAY 3 3 modules

Gemnoor Tech
PRIVATE LIMITED
BUILD
For more info: https://gemnoor.com

MODULE 7

Evidence-based verification and closure

- Ingesting a second scan or fixture; fixed, mitigated, not reproduced and unknown states
- A finding is never resolved just because a ticket is closed
- Audit history of state transitions

HANDS-ON LAB
Write integration tests that prove closure requires fresh evidence or a documented exception.

DELIVERABLE
Verification workflow.

MODULE 8

Metrics, evaluation and dashboards

- Duplicate reduction, ranking quality, unsupported-claim rate, ticket quality, verification correctness
- At least one false-positive and false-negative review
- Metrics dashboard and audit

HANDS-ON LAB
Run the evaluation set and publish the dashboard.

DELIVERABLE
Evaluation report.

MODULE 9

Capstone: end-to-end vulnerability workflow

- Scanner export to verified closure in one repeatable run
- Acceptance criteria, limitations, demo

HANDS-ON LAB
Capstone build and demo.

DELIVERABLE
End-to-end vulnerability management lab.

Capstone

Capstone project

An orchestration layer that imports scanner findings, deduplicates them, enriches them with asset and exploit context, explains priorities, raises idempotent tickets and verifies closure from fresh evidence.

Gennoor Tech · LEARN · INNOVATE · BUILD · For more info https://gennoor.com

Assessment

Assessment area	Weight	Evidence
Working functionality	25%	Demonstrable end-to-end workflow in a repeatable lab
Security and safety controls	20%	Threat model, authorization boundaries, approval gates, safe failure
Testing and evaluation	20%	Automated tests, golden dataset, measured quality and known limitations
Engineering quality	15%	Clear interfaces, error handling, reproducible setup, maintainable code
Evidence and explainability	10%	Traceable findings, sources, assumptions, logs
Documentation and presentation	10%	Architecture diagram, README, runbook, demo, lessons learned

Engineering principles used throughout

- Model output is a proposal, not a fact: every finding and generated change is validated by deterministic checks and tests.
- Authorization lives outside the model: tool access is enforced by application code, scoped credentials and policy.
- Human approval for consequential actions: deploying a mitigation, changing production configuration, rotating credentials.
- Decisions, evidence, model and tool versions, approvals and outcomes are logged without recording secrets.
- A model adapter and a tool adapter layer keep every build provider-agnostic and enterprise-portable.
- Quality is measured: false positives, false negatives, reliability, latency, cost and safe failure, not just a demo.

© Gennoor Tech · Train. Innovate. Build.